ActInf Textbook Group ~ Cohort 4 ~ Meeting 18 (Chapter 8 part 2)

TextbookGroup/ParrPezzuloFriston2022/Cohort_4 | Cohort 4 ~ Meeting 18 (Chapter 8 part 2) | 57:03

all right hey everyone it's October 24th
23 and we're in our second discussion of
chapter 8 in cohort
4 so we can do any number of things but
does anyone want to begin with
any thoughts
on chapter
8
we can look over past notes we can look
at the extent
questions we can hear any new questions
we can look at the
text
yeah we'll have a very simple question
uh in in the continuous case is there
some kind of standard code to use to try
modeling I mean to test some very simple
models uh as there is for the discret
case with you know the
SPM
framework a very good question
um I'll give my thought just it may not
be fully accurate
in the mat lab SPM based and python pmdp
based code there's been an emphasis on
the discret
modeling in the RX and fur Julia
package I believe that there's a lot
more continuous time
models okay so I'm not offand familiar
with
[Music]
um any continuous
time uh implementations we can like look
at the implementations table that we

have eventually I'd love to
see um the these repositories of course
expand and grow but also
annotated with different features of the
model so then we could filter and say
I'm looking for this language continuous
time hierarchical
model
but almost all of the Python that I've
seen was
discreet
okay any other random thoughts or or
question questions otherwise we can look
at some some popular
questions and look at the
notes does anyone have
any just even for outside of active
inference thoughts on continuous and
discreet
time anything with digital and
analog do do we see one of them as being
the general case of the
other or do they both derive from a
point
one oh yes yes Daniel um I'm just
wondering what is fundamental difference
between um continuous time model and
discret time model is that is that the
fundamental difference is in terms of
the um the state transition
uh like in in the discret in the
continuous model is represented as a
differential
equation and in the dynamic model in in
the discrete model is expressed as a as
a kind of a difference
uh
question

does anyone have a thought on
this yeah here what does continuous mean
continuous in time or continuous in the
in the uh State
space uh or refer to
both I suppose it you know most means it
means the continuous in time right not
um does anyone want to give a thought on
this yes it is referring to the
treatment of
time so here in figure 43 We have
basically chapter seven and chapter
eight laid
out in the discret time setting
then let's talk about what's the same
first broadly their architecture is the
same that's why they're being laid out
this way what else is the same priors
are set the same
way similarly enough D and also the
upstairs the policy selection
apparatus is broadly the same we're
still dealing with free energy based
policy selection
mechanics um also what's the same is the
hidden state to observable mapping the
partially observable setting with two
so really what's different is kind of
like the core of the Horizon here in the
discrete time setting we have
explicit hidden State um or external
State um calculations at given time
points T minus one t t+
one right those themselves as you
pointed out could be
continuous so that could be like a
number between zero and
one um but we're estimating that at 1

p.m 2 p.m 3 P.M 4
P.M right so and then um in the
continuous time
setting we only are keeping an
explicit
prediction of the Hidden State at the
current moments
X and then we're dealing with um the
past and the
future with a tailor series
expansion also known as the generalized
coordinates so here whereas B plays the
role of a marov transition Matrix in the
discrete time model and has a kind of
straightforward interpretation where
like you have the hidden State at a
given time you multiply it by the B and
then like you basically transition into
the next
state right the analogous role in The
Continuous time setting is a temporal
derivative right and so we're extending
out a tailor series which technically
goes forever uh usually the accuracy of
a tailor series exper ion begins to drop
off quite rapidly unless it's a
repetitive
signal
um so and then in the continuous time
setting I guess you could have a
discreet State
space like you know the light switch is
either on or
off and we're going to have a a model
that's continuous in time of being on or
off
that doesn't mean it's the best way but
it's definitely a good point that you're

raising which is that the treatment of
time has a um
very strong bearing
on the semantics of the generative model
whereas whether a given variable like an
observation or a hidden
state is itself continuous or just
street is a little bit of a smaller
difference this this figure
4.3 okay yeah
um
um yeah question continue
sorry one are you are you done no no no
go ahead go ahead okay uh
yeah so uh to go on on this uh in the
discrete time um in the
implementation uh within each step there
is some kind of continuous updating of
the neural activity associated with all
the
states
um
so uh you know how does that uh go in
this picture
this
yeah this is also a very good
question like the the this kind of
interpretative yeah
continuous
dynamical
component that seems to
be
extraneous
or it's unclear if
it's purely
interpolative
like if a calculation is being done here
here here and then it just

being like kind of like spline
fit or rather if there is
a continuous time differential equation
or
[Music]
something but then if there was a um
exact
concordance of the discrete
time and
some differential equation solving
underneath why wouldn't we just focus on
that
concordance I don't know if this is
accessory code inside of
SPM uh I mean that's linked to the to
the message passing within each
step So within each step you know if you
have for example uh
I mean usually they use 16 steps uh to
uh implement the message passing uh and
you will have this uh differential
equation implemented has uh you know 16
different uh updates in a
discretization of
uh okay so then would we say that this
is truly a continuous time interpolation
or is it just that we're doing like
16 with we're we're treating each time
step in the discret model as like in
Epoch and then having 16 kind of
adjustment steps
within yeah the later is my the letter
is my understanding
yeah yeah so it could
be so the there well
neurobiologically they analogize this to
the local field potentials which is like
a EEG recording but just statistically

the of change of log
expectations
so using message
[Music]
passing they can continue to run message
passing um okay well let's let's assume
this the trivial case which is the
variable is already exactly update it's
already at its stationary state so new
irrelevant information comes
in um then
whether you message passed one time or a
million times it wouldn't really matter
right in contrast there might be
situation which is going to be more
prevalent where there is some amount of
basian updating that needs to
occur
um the whole point of using free energy
is that we're able to
approach problems that we can't do one-
step exact Bays on
we have something that's incrementally
optimizable through message passing with
free energy
minimization so then how
many rounds of message passing do you
have to
do well one approach is when the um rate
of
change of free
energy um or the rate of change of log
expectations is is
low um or sometimes I've seen when like
the overall Delta is small like just
when round after round we're getting
small updates but
but both of those H have a degree of

freedom because maybe you could continue
in that slow regime for a long time just
like any uh
model so they're doing more multiple
message passing
rounds have you run the code to to
reproduce
7.13 uh not this one no okay yeah I
think that would be helpful to see
because other even
still like does it message pass up to
here and then it goes down to there and
is that point two or is
two well I would uh
I would say that for each I mean each
step is composed of three steps here uh
and for each small step you have this uh
you know 16 by default uh message
passing did they say the 16 in the text
or is that from the
paper uh well that's uh the default
implementation with SPM okay about the
text here yeah sorry okay
yes
interesting I I think also this is a uh
a place to look at the RX and
fur um
toolkit um as well as the reactive
message
passing um
variant so in the SPM based message
passing every every message EV basically
one round is Advanced every single node
is messages are propagated across the
entire network um in the reactive
message passing setting it opens the
door to like adjust in time type
um uh operation where different nodes

can be receiving messages at different
rates in fact later today um we have
live stream 55.0
on message passing and then in the
coming two weeks um with Magnus kudal on
message passing um but it makes me
wonder
like is there still message
passing in the continuous time
model I'm tempted to say yes any Bas
graph since any B Gra
has a message passing
duel then whatever the semantics of the
base graph are whether it's describing
Markov transition probabilities or
whether it's describing a temporal
derivative for Taylor series
expansion you still could construct the
message
passing to do
that
[Music]
um I I I I'm actually even going
to bring in this one slide for message
passing just
because it
uh
okay three kinds of
graphs I and interestingly all three of
these kinds of graphs come into play in
the textbook but I think in this 2017
paper they they lay it out most clearly
basy and graphs nodes are variables
edges are
dependencies figure 4.3 base
graph for Factor
graph nodes are
functions of a distribution over

variables edges represent variables per
se
so here's 4.3 well here's figure 7.3
same thing
4.3 so these two are
equivalent now this which is to say
4.3 is equivalent to this forny Factor
graph and then the Third Kind of graph
is the neural
network nodes are sufficient statistics
edges are exchanges of sufficient
statistics so it's kind of interesting
on one hand we shouldn't be super
surprised because the graph is obviously
an extremely
General type of
data so the idea that there would be
like multiple kinds of
graphs it's not too
surprising
can you give an example of the neuron
Network
gra which one is neur
Network um to the one we just showed on
page 26 there are on page six you'll
have you'll have basian graph uh and the
uh Vector graph
right yeah so here on the top are the
base
graphs then okay so these two are
literally identical this is from the
textbook
hash Tex group and this is from the
graphical brain paper which I'll just
put into the chat just in in case you
want to see
um then every base
graph any bayy network can be expressed

as a factor graph right so
basically they
contain the same NE same necessary and
sufficient information however there are
operations and procedures that can be
done on a factor graph that cannot be
done on the basian graph
representation so under the hood which
is I think something to investigate in
SPM and Python and
Julia when we specify our generative
model as a base
graph and then we use standardized
routines for message passing it's very
interesting to ask well what's happening
under the hood that converts that base
graph into a factor
graph the advantage to the factor graph
key Advantage is that as these numbers
suggest you can create node
local procedural
Logistics so you can develop per
node an order of operations that
calculates that
node which allows you to develop um a
procedure to to accurately update the
entire
graph whereas in if this base graph is
presented it's kind of like well you
know should we should we calculate this
part first or should we calculate that
part first is it even going to matter um
okay now the Third Kind of
graph the neural
network or
circuit again this is this is I think
worthwhile to go into because I think
when they're talking about a neural

network this could this could be wrong
but
um these are the kind
of yeah let's let's look in graphical
brain just to see if they give an
example the neuron netor graph yeah yeah
because I'm not sure if they're okay
yeah okay we'll get okay here we
go here's a neural Network graph so I
don't know if this
includes standard SL mainstream neural
network architecture like conet type
stuff I'm I'm not sure if it would
include
that um and then also we've seen neural
cognitive architectural
representations described as a base
graph for
example
in let's say here chapter 5 well for so
that's figure 52 predictive
coding so here
the these kinds of things now on one
hand I see I see that in terms of um
sufficient statistics of unknown
variables and other auxilary variables
like prediction errors that's literally
What's Happening Here the mean is the
summary statistic of the central
tendency and then the um error
residual but I'm not sure if this can be
interpreted both as a base graph and a
neural graph or if this is one or the
other because in either case it's
slightly different in its
representation one key piece that I
think is happening here is there's like
multiple at each

layer but I'm not sure if that's being
if these threes or the two at each are
being used to describe three time points
kind of as traditional or whether it's
alluding to the idea that there's a
population of
neurons and that the variable is a
summary
statistic on like a rate
code
should not be taken too
seriously yeah I'm just wondering if if
they are kind of have oneto one
correspondence uh among the three uh
graphs yeah let's that would be a great
like that would be like the triple
play with the three kinds of graphs
whereas they only really show two in
connection um figure
11 figure
nine
what but here's what figure 10 looks
like so
but this looks a lot like
figure
5.5 right we have the yeah
prefrontal um the sort of
abstraction um goal
selection
then the descending
prediction down into the
dopaminergic policy
selection we talked about the tradeoff
between basically habit driven policy
selection on the left MH and then
expected free energy sharpened policy
updating on the right that's like the
dopaminergic

tone and
then the descending prediction down to
the um spinal Arc with a kind of simple
differential
based um set point predictor Sentry
motor
actuator
um this is like
literally the same
we have that prefrontal
stack here we have a um policy and a
g and then um ultimately a
descent into the spinal
Arc but what makes
this question one what makes this
different than just saying well it's a
Bas network with a neural
semantics I mean don't the nodes as
unknown variables
already have an interpretation as
sufficient
Statistics question one question two oh
yeah go
ahead yeah one thing is that's you know
when when you have categor categorical
distributions to get the sufficient
statistic you need the full
distribution uh where you in continuous
time I think they always work with um
Goan
distribution you know in which you know
having the mean and standard deviation
is
enough yeah they're they're definitely
working calmly with the Gan family
yeah although in the hierarchical
gussian setting the hierarchical
gaussian

filter has high expressivity so it's not
like they can only model Simple Things
No No Yeah Yeah
Yeah so how what what makes this a
neural
network are they using neural network
like many machine learning researchers
would use neural network today sigmoid
activation function all that I I don't
EX exactly expect so
um so how is this different than a Basi
Network part one and then are are these
models in cont could these models be
simply does not include continuous
message
passing so here they're dealing with
categorical and continuous State
spaces
M oh see but
here these generalized observations
describe a trajectory in continuous
time so here we see the X X Prime X
Prime so this is a a lot like the
derivatives over time like first order
second order over time yeah okay and and
and those are the generalized
coordinates of motion you know the
position of the car the the um velocity
of the car acceleration of the car and
so on and then you know the more
derivatives that you have that's using
the tailor series or the generalized
coordinates to expand out um live stream
26 on the uh basian Mechanics for uh
stationary processes or something like
that that
um that is very helpful like let's just
say you had a pendulum just moving back

and forth well its position is obviously
never the same moment to moment it's
changing its velocity is changing moment
to
moment but you're going to get to a
derivative not that high up for a
pendulum
where that derivative is not
changing and that is stationarity on the
generalized coordinates or if you had
something that was moving around in a
circle again the position is changing
the velocity would always be changing
like the actual Vector heading would be
different at each moment but the Accel
oh it's accelerating a little bit
forward and to the left it's going
counterclockwise so that would be
stationary
so that's a huge advantage of the
continuous time is like if you can find
the
articulation where something has a
regular
mechanics
then the the generalized coordinates the
Taylor series
expansion like capture it really really
really
concisely
whereas if you were doing a discret
model of that something going around in
a
circle you you would I don't know maybe
struggle to to pull out that
Simplicity especially
if huh yeah I'm thinking when you
implement this uh continuous time model

in Compu computer you have to discretize
it the time anyway
right yeah this is a great
question well it then becomes a discret
time if discret the time will it become
a you know discret uh model or it's it's
actually fundamentally different from
the discret
model you're you're right I would
say that still would not be called a
discrete time model because discrete
time model the B Matrix would be you
know the Markoff transition
probabilities whereas here you're still
trying to learn the
derivative so then you take some
approach like you have some delta
T and then you shrink the delta T and if
you find that as you're shrinking the
delta
T you're converging on your estimate
of the
derivative you know we're studying the
line Y equals X and we we start with a
with a Delta X of five and we find the
slope is one and then we do Delta x
equals four it's still slope of one
Delta X so we're converged in our
estimate of slope as we shrink the
Delta so then we feel confident in our
discretization but we're still in the
continuous time setting
but we're just needing to take this this
digital
approximation on the continuous on on on
estimating the
um um continuous time
Dynamics Lance D

Costa is is one of the researchers who
has I
think expanded on this some of the most
because
in in
26 basing Mechanics for stationary
processes here's where there was the
discussion of the non of non-stationary
steady States stationary non-stationary
steady states in the generalized
coordinate settings also even with three
layers like x x Prime X double Prime
that corresponds to PID control in
engineering the um something integral
derivative um so this is a very common
engineering um technique
then um in
52 this question of the delta
T came into play so like obviously we
want the underlying free energy
landscape to be
smooth because we want to be like
analytically well behaved we want to be
able to take
derivatives so we want the free energy
landscape to be
smooth but we're doing it on digital
computers so there has to be some
approximation
step in in space and
time even if we're dealing conceptually
with a analytical space that is
continuous in space continuous in time
um
so how does that work
well in this this paper they talk about
how do you do inference on that um on
the delta

T how do you decide how much that step
should be and there's a very evocative
um
image so we're we're the ball is rolling
to the bottom of the hill we always talk
about this but we're on a computer
simulating the ball rolling to the
bottom of the hill
so um we are going to make these
discreditation
now how much time should we simulate
okay we have the equations that say that
the ball is going to roll smoothly in
time smoothly in space but we need to
simulate
that they talk about this um delta T
question in terms of accelerated
optimization like you could choose a
delta T that was so ridiculously
small that it basically you know it
wasn't
moving like it was just super super you
were just modifying a parameter by like
a
billionth um conversely you could make
delta T that would lead to these kind of
like chaotic
sampling because you'd be like you'd be
like okay the ball's here and it's
headed this way and then you just go
okay now let's just put it over there
and then now it's like a totally
different space and then it's like whoa
what is happening now it's going that
way so if your jump is too
big then you're you're like you're
getting kind of
like I don't know if You' say you're

getting Alias on that space But if
you're taking too large of jumps you're
not actually going to detect the
regularities landscape you'd be on the
side of the bull and you'd overshoot and
you'd find yourself way off the curve
you'd have to recorrect back to the
curve and then you'd find yourself going
way down here and so then there's this
kind of second order um inference
question about what the delta T should
be and then they talk about like um that
in terms of the ball rolling to the
bottom of the hill with different fluid
viscosities so if you're like in honey
you stay kind of move you never really
reach a terminal acceleration
velocity versus if you're like in
water versus
Air you like gain more
speed and so that that
leverages the position of the ball on
the bull and the vector
Direction but then you can you can
knowing how far you should
go is as a higher derivative question
it's like if the car is going one mile
an hour
then there's going to be a delta T
that's going to lead to Optimal sampling
of that car like one second later but
then if the car is going a thousand
miles an hour you might want the delta T
to be
shorter do all of these
like again I think this um again returns
to the code
question

um do all do we just like swap
out continuous time if we have a model
with a given State
space how hard is it to go from a
continuous to a discrete
time
implementation like is it could we just
flag it up at the top and just say we're
going to do both then we're just going
to see what they look like or you know
is it is it a flag or or a setting that
can just be called or does it require
more of a
structural Custom Creation for one or
the other such that it's a little bit
harder to
repurpose I don't know in fact I've done
very little with the continuous time
models but the RX and fur the RX and
fur it
it
certainly is is
[Music]
um it certainly is is fine with with
continuous so so this is a package for
continuous uh time
models um it's actually a a package for
um like message passing and basy graphs
generally so you can make you don't this
isn't even an active inference specific
package and so I think some of the
reasons why pmdp gets more use at least
today first off more people know python
than
Julia second off
pmdp like
SPM has multiple
methods that are specifically relevant

for doing active inference generative
models like there's a
class to define the the PDP agents
so that makes it a lot easier to
Define whereas
here here's you can do active
inference
but
[Music]
um you have to write it from scratch
basically
but so it's all there
because if you if you can frame it as a
basian graph and we know that we
can then RX
infer is a general Bas graph
implementation approach which allows you
to specify the base
graph and then under the
hood I
believe flips it into the message
passing format and actually does the
inference with message
passing but these are these are
very so
yeah saying I'm not a Julia expert
either but like some features that are
presented already in RX and
fur are very
promising let's see if there's
continuous time specifically yeah good
try out SPM the discret case I tried out
it's it's quite nice using the SPM
package oh yeah
totally I haven't tried any other
packages yeah I think this could be a
good I mean Bert Dev rise and again
we're having live stream in the coming

weeks with the authors of of this
package and so we can ask these kinds of
questions
[Music]
um but just in I mean let's
like this
textbook the the 2022 textbook that were
ostensibly
discussing um is kind of the Matt lab
SPM era and
approach yes
um pmdp represents a sort of partially
first principles approach partially
ported over from SPM
approach and they focused their
development on discreete time
modeling what is the name of
pmdp pmdp yeah okay infer actively the
pmdp package um whereas RX and fur set
out to
solve a significantly more General
problem of just doing basian
inference but using pmdp to construct
arbitrary basian graphs is probably not
the best package for that but again it
has all these helpful methods for
making
pdps um move movement is a classic case
of continuous time
modeling the analog realtime continuous
nature of proprioception and actuation
so sensory motor control this is an
absolutely classic setting for
continuous time
modeling by thinking
about what that con continuous time
model is in the game of
doing not

as um maximizing utility a reward
function per se but reducing
Divergence to a set
point and finding paths or trajectories
in a space towards a set
point already this
foreshadows this kind of like sensory
motor continuous layer and then a higher
order
discret now could be discrete time but
at the very least discrete State space
with like selection of different
goals so you know to oversimplify the
brain is send is is um setting discret
preferred States for the body to be in
and then the
body uses its um basil intelligence to
realize paths navigate paths to that set
point there is a discussion I think we
may have talked last week about this sum
about the sensory
attenuation about the Precision Dynamics
and we talked about like getting up out
of a chair we talked about moving with
the eyes the icade and about
how as during the execution of a
movement you're still getting sensory
feedback and so if you expect the the
consequences of a movement like when I
when I um move my eyes or move my head
I'm expecting all of the visual input to
change a certain way when I get out of a
chair I'm expecting my um touch
receptors to feel a certain way those
expectations are calibrated out so that
they do not rise to our
attention and so that can be understood
in terms of a transient suppression of

attention which is
attenuation and then once the body is
fixed again or the Gaze is fixed again
there's a re-engagement of precision so
you know pay attention to when things
outside of your control are
happening but calibrate
out the expected sensory consequences of
action and the easiest way if you know
that there's going to be like um you
know if you know that you're going to
drop something loud
just turn off your hearing for a split
second while you drop something
loud instead of engaging in in some more
sophisticated like noise
cancellation
method they then move to an ecological
analogy with a generalized lock of
voltera Dynamics this is also called
winterless
competition because like plants
herbivores carnivores
like none of them can like wa in they're
all in this
oscillatory relationship in this
three-dimensional
space and that's also been used to model
winterless neural Dynamics like if you
have two brain regions that have a
mutual inhibitory relationship like the
one that's on top inhibits the other
one
um until the rep the repression switch
happens so that's explicitly brought
into this uh winterless competition lock
of voltera was some of the earlier
models of sequential actions in active

inference and this Thomas Parr in the
bookst Stream 2.1 I think he talked
about this so like there was sort of the
continuous time this was like you know
2008
2012 The Continuous time models were
great with like real time
flow but doing like a
um so it was all good for you know the
arm close the arm okay now open the arm
now close the arm but to do a
multi-stage action how do you get that
into continuous time so the first
approach that was shown here was um like
saying how do you get a multi-stage
action where first I want the plant
population to drop then I want this
carnivore population to go up or
something so there are ways to tune The
Continuous relationship so that you can
get um a kind of canonical
unrolling in response to a certain
stimuli or endogenously generated which
is what they did in the handwriting case
however as Thomas pointed to people
wanted to account for explicit planning
because even though the models tuned up
to do this kind of repetitive scribble
there was no explicit
planning and so that motivated
the development of the discreet time
formalism and the PDP formalism which
had been widely used for
planning and now we're kind of merging
back those streams with the continuous
and discreet times especially in light
of the recent advances in basing
mechanics with path

integrals G Theory these kinds of topics
learning and continuous
models generalized synchrony now
generalized synchrony you could also
think about this in a discreet time
setting
but it works really nicely in a
continuous time setting
too this was
um this specifically coupled
Loren's
system
explored more in live stream 34 on the
stochastic chaos and marov blankets
turns out that you can get generalized
synchrony
between um two chaotic systems between
two non-chaotic systems and so on so you
can get entrainment and generalized
synchrony even when there's an
analytical chaotic
relationship
is it hard to fit chaotic processes in
real life yes
probably but in the example of the two
birds that are like
singing where their turn taking is
described by like it's like you're like
spiraling on one side of the Larn
attractor and then there's like a flip
into the other side of the phase
attractor that describes the turn taking
and the generalized
synchrony
well and combining the generaliz
synchrony plus learning with the two
birds we can look at how
like through when we have a learning if

the models didn't have learning
obviously they would just continue to
act along wherever however they were in
the beginning but with learning they can
come to have higher Mutual information
on each other closer to this like Y
equals X line so that represents the the
tightening of of the generalized
synchrony which doesn't mean lock step
doesn't mean that they're seeing at the
same time it actually means that they're
turn taking is sharper in its
alternation then in the close missing
line here in the close of the
chapter they introduce the kind
of folk psychological model that gets
explored in live stream 46
that we discussed earlier where we have
a continuous time sensory motor
actuation as a leaf in a higher order
discreete time
model then there are these just
[Music]
um examples of icting with mixed or
Hybrid
models where we have a discrete time
goal um selection and then the
continuous time
um
Cate daian mixture
model big uh topic here about iterative
clustering kind of like K means
clustering and then they
review 10 or 12 continuous
time
settings
so a very interesting one certainly one
that we could benefit a lot

from having like the same setting
described with continuous and discreet
time to get a better handle
on whether we do we make the state space
and then do we add the time layer and we
can do it both
is there any downside to making models
that way or if we think back to chapter
six and the
recipe is the decision that we make
about continuous or discreete
time something that happens very early
and in a more committal way I don't
know and the way that the packages are
today may not be how packages are
forever so then I have one question
regarding the the example giv in qu giv
in chapter eight are this example like
other figures are they are simulated by
computers right they are not the data
connect from from experiment or from uh
from the nature
right correct in the examples here the
generative model was specified and then
synthetic data were generated so that's
kind of like the forward Direction and
then chapter nine is where we get into
the discussion of okay now what if we
have empirical data and we want to
parameterize the generative model but
here they just set the generative model
and then synthesized
data however they obviously set up the
generative model to anticipate the kind
of data that they were collecting in the
lab so this is the forward Direction and
then chapter 9 is when we go into
modelbased Data

analysis with going from empirical data
to parameterized generative
models cool
okay thank you Daniel thanks thank you
thank youil thank you on Sean farewell
bye thanks bye